UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages

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QUESTION 1

a) Solve the inequality 2 - 4x < -6. Express the answer in interval notation.

(4 marks)

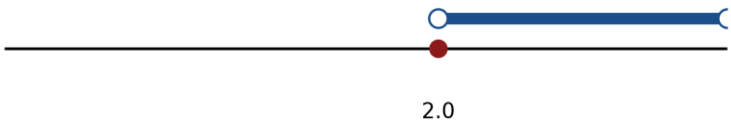


Figure: solution on a number line.

b) Solve the inequality |5x + 2| < 3. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Sketch y = 2|x - 3| + 5. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

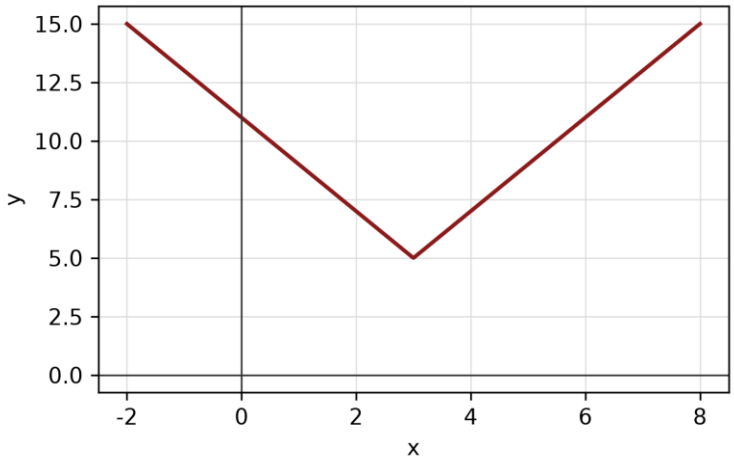


Figure: absolute-value graph.

QUESTION 2

a) Given f(x) = 2x - 6 and g(x) = -5/(x - 5), find (g ∘ f)(x) and state its domain.

(5 marks)

b) Given f(x) = (1 - 4x)/(x + 3) and g(x) = √(3x + 3), state the domain of each function.

(4 marks)

c) Find the equation of the straight line passing through (1, -3) and (5, 0).

(4 marks)

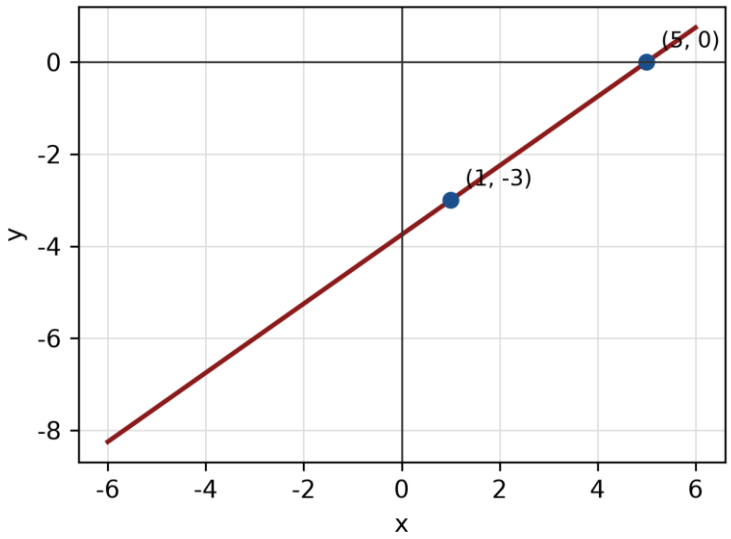


Figure: line through the two points.

QUESTION 3

a) Given $z_1 = 2 - 4i$ and $z_2 = 5 + i$, find $z_1 - z_2$ in the form $a + bi$.

(4 marks)

b) Evaluate i^33 and simplify completely.

(3 marks)

c) Solve the logarithmic equation log_2(x) + log_2(x - 14) = 5.

(5 marks)

QUESTION 4

a) Solve the simultaneous equations

$y = x^2 - 4$

$y = x + 8$.

(5 marks)

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b) Write a system of inequalities that describes the shaded triangular region with vertices (0, 0), (4, 0) and (0, 5).

(5 marks)

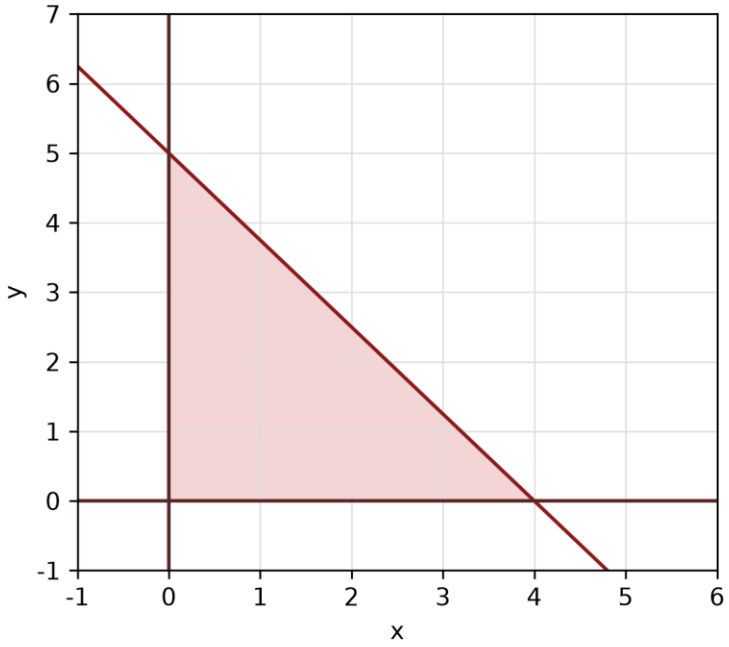


Figure: shaded triangular region.

c) Write the equation of the circle with centre (-4, -2) and radius 4. Hence write its general equation and sketch the circle.

(5 marks)

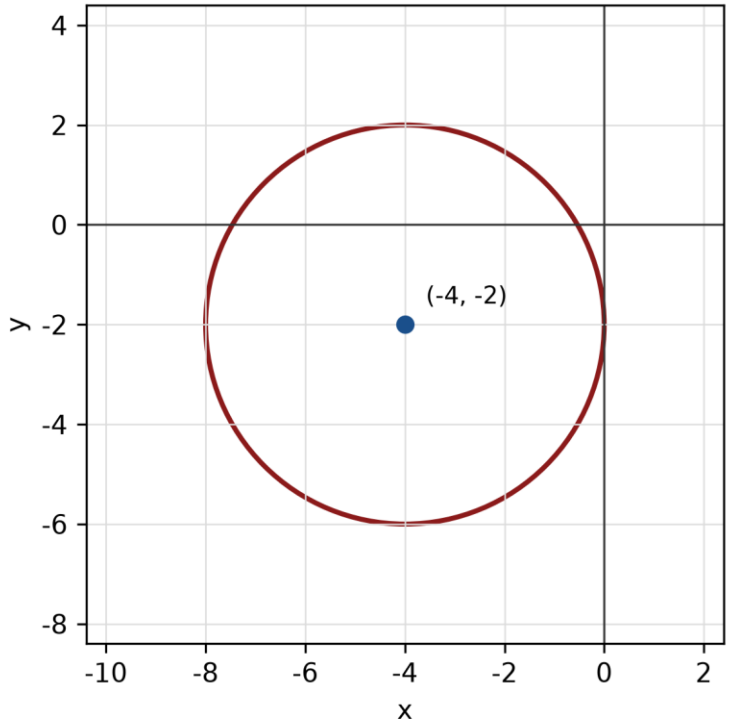


Figure: circle.

QUESTION 5

a) Prove the identity $\sin^2 x/(1 - \cos x) = 1 + \cos x$.

(5 marks)

b) Find the perimeter of the sector with radius 9 cm and central angle 120°.

(4 marks)

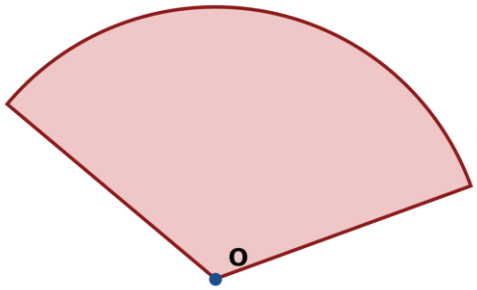


Figure: sector.

c) A right triangle has legs 9 cm and 8 cm. A sector of radius 3 cm and angle 90° is cut from it. Find the area of the remaining shaded region.

(5 marks)

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$
$\sin(A + B) = \sin A \cos B + \cos A \sin B$
$\cos(A + B) = \cos A \cos B - \sin A \sin B$
$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$
$\cos 2A = \cos^2 A - \sin^2 A$
$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$
$\sin 2A = 2 \sin A \cos A$
$\tan^2 A + 1 = \sec^2 A$
$\tan 2A = 2 \tan A / (1 - \tan^2 A)$
Arc length, $s = r\theta$
Area of sector = $\frac{1}{2}r^2\theta$
$a / \sin A = b / \sin B = c / \sin C$
$a^2 = b^2 + c^2 - 2bc \cos A$
Area of triangle = $\frac{1}{2}ab \sin C$

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